

Technical Research Report

Reproducing Paper 3 and Evaluating the Impact of Data Duplication on Model Evaluation and Generalization

Course / Assessment: 11.1HDTask— Machine Learning Research Report

Student Name: Adel Saleh M Althagafi

Student ID: s223817705

Date: September 2026

Video Presentation Link :

[https://drive.google.com/file/d/1VwMNAxbwwftOgQ3kG_wVX21kZ9_fvv9m/view?usp=sharing]

Code & Data Repository Link (GitHub):

[<https://github.com/deepmind00/Heart-Attack-Prediction-Reproduction-Audit>]

Submitted archive contains: complete source notebooks (.ipynb) · original dataset (heart.csv) · serialised ML artefacts (.joblib) · experimental output tables (.csv) · README.md with reproduction instructions.

Abstract

This study presents a reproduction of the machine learning methodology reported in Paper 3, "*An efficient stacking-based ensemble technique for early heart attack prediction*," together with an empirical investigation of data duplication and train-test contamination. The reported methodology was reproduced using the complete 1,025-record heart disease dataset. The reproduction produced unusually high predictive performance, although the exact numerical results differed from those reported in the original paper. Random Forest, XGBoost, and the proposed Stacking Ensemble achieved 100.00% accuracy and an AUC of 1.0000, compared with the published Stacking Ensemble performance of 98.53% accuracy and 0.9880 AUC. To investigate the stability and validity of these results, an independent exact-record deduplication analysis was subsequently performed. The original dataset contained 723 duplicate observations, leaving 302 unique observations after exact-record deduplication. Re-evaluation on this deduplicated cohort substantially changed the observed model performance. The Stacking Ensemble decreased to 75.41% accuracy and 0.8626 AUC, while Logistic Regression achieved the highest accuracy of 80.33%. Gaussian Naive Bayes achieved the highest AUC of 0.8842. An additional duplicate index-overlap analysis found that 202 out of 205 test observations in the original 1,025-record experiment had exact duplicate counterparts in the training partition, corresponding to 98.54% of the test set. This provides direct empirical evidence of train-test contamination caused by duplicate observations and offers a strong methodological explanation for the unusually optimistic performance observed on the full dataset. The findings demonstrate the importance of duplicate auditing, appropriate data partitioning, and independent validation when evaluating machine learning models for medical prediction tasks.

1. Introduction

Machine learning applications in cardiovascular disease prediction require reliable evaluation procedures to ensure that reported performance reflects genuine generalisation rather than characteristics of the dataset or evaluation procedure.

Paper 3 proposed a Stacking Ensemble architecture for early heart attack prediction using a 1,025-record heart disease dataset compiled from multiple UCI sources. The authors reported a Stacking Ensemble accuracy of 98.53% and an AUC of 0.9880.

Although these results are highly promising, unusually high predictive performance in medical machine learning warrants careful examination of dataset construction, duplicate observations, train-test separation, and evaluation methodology. This report therefore investigates the paper through two complementary experiments:

Experiment 1 — Paper Methodology Reproduction: Reproduction of the reported machine learning pipeline using the complete 1,025-record dataset.

Experiment 2 — Deduplication and Generalisation Analysis: Re-evaluation of the same model suite using an exact-record deduplicated dataset containing 302 unique observations.

In addition, a duplicate index-overlap analysis was performed to determine how many test observations had exact duplicate counterparts in the training partition. The objective is not only to reproduce the reported methodology but also to assess how duplicate observations can affect model evaluation and the interpretation of generalisation performance.

2. Dataset Architecture and Exploratory Data Analysis

The dataset contains 13 predictor variables and one binary target variable. The target variable is encoded as: **0** = No disease, **1** = Disease present.

2.1 Feature Specifications

<i>Feature</i>	<i>Description</i>
age	Age in years
sex	Biological sex (1 = male, 0 = female)
cp	Chest pain type
trestbps	Resting blood pressure (mm Hg on admission)
chol	Serum cholesterol (mg/dl)
fbs	Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)
restecg	Resting electrocardiographic results
thalach	Maximum heart rate achieved
exang	Exercise-induced angina (1 = yes, 0 = no)
oldpeak	ST depression induced by exercise relative to rest
slope	Slope of the peak exercise ST segment
ca	Number of major vessels (0–3) coloured by fluoroscopy
thal	Thalassaemia classification

2.2 Cohort Diagnostics

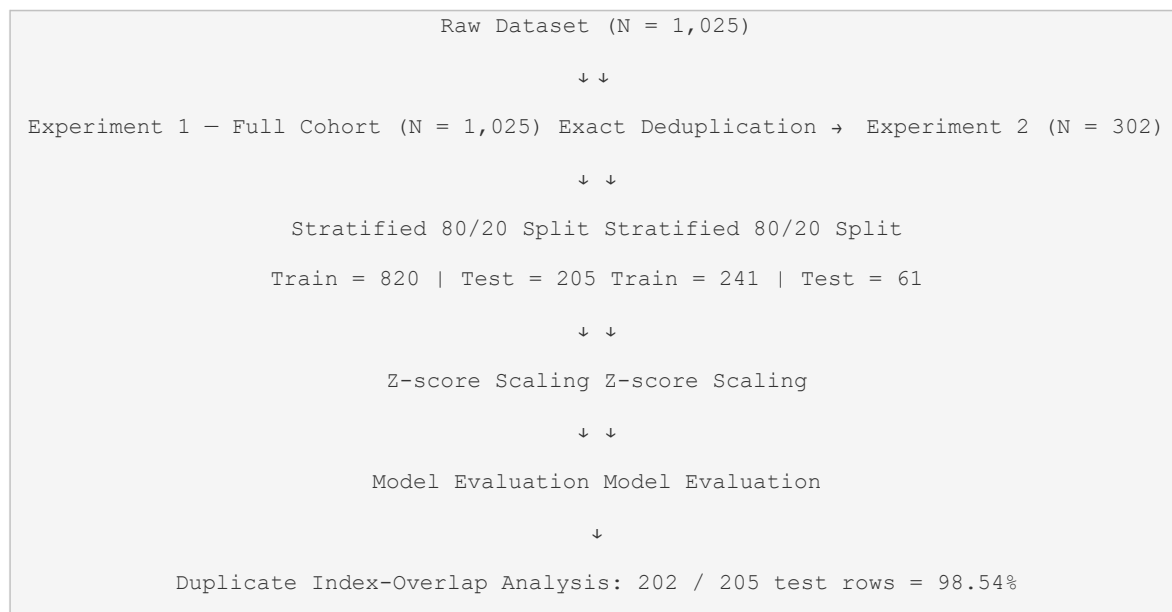
Diagnostic

	<i>Result</i>
Total observations	1,025
Total variables	14
Missing values	0
Positive observations	526 (51.3%)
Negative observations	499 (48.7%)
Duplicate observations	723
Unique observations after deduplication	302

The exact-record deduplication process reduced the dataset from 1,025 observations to 302 unique observations. The deduplicated cohort comprised 165 positive cases (54.6%) and 137 negative cases (45.4%), maintaining a reasonably balanced class distribution.

3. Methodology and Experimental Framework

The experimental framework used the original dataset as the starting point and created two evaluation tracks, as summarised below.



3.1 Preprocessing and Validation Controls

The following controls were applied consistently across both experiments: Global Random Seed = 42; Stratified 80/20 train-test split; Z-score feature standardisation fitted on the training partition only; Evaluation metrics: Accuracy, Precision, Recall, F1 Score, and AUC.

The standardisation transformation applied was:

$$z = (x - \mu_{\text{train}}) / \sigma_{\text{train}}$$

where x is the original feature value, μ_{train} is the training-set mean, and σ_{train} is the training-set standard deviation. Fitting the scaler only on training data prevents the test-set distribution from influencing the preprocessing stage.

3.2 Algorithmic Suite and Configuration

Model	Configuration
Logistic Regression	max_iter = 1000, solver = lbfgs
Decision Tree	criterion = gini, splitter = best
Random Forest	n_estimators = 100, criterion = gini

<i>Model</i>	<i>Configuration</i>
XGBoost	n_estimators = 100, learning_rate = 0.1
Gaussian Naive Bayes	Default prior estimation
K-Nearest Neighbors	K = 5, Minkowski distance, p = 2

Stacking Ensemble — Base estimators: Decision Tree, Random Forest, XGBoost, Gaussian Naive Bayes, K-Nearest Neighbors. Meta-learner: Logistic Regression. Cross-validation: 5-fold stratified.

4. Experimental Results

4.1 Experiment 1 — Paper Methodology Reproduction (N = 1,025)

The first experiment used all 1,025 observations without removing duplicate records. The reproduction produced extremely high performance for the tree-based models and the Stacking Ensemble.

<i>Model</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F1 Score</i>	<i>AUC</i>
Logistic Regression	0.8098	0.7619	0.9143	0.8312	0.9298
Decision Tree	0.9854	1.0000	0.9714	0.9855	0.9857
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000
Gaussian Naive Bayes	0.8293	0.8070	0.8762	0.8402	0.9043
K-Nearest Neighbors	0.8634	0.8738	0.8571	0.8654	0.9629
Stacking Ensemble	1.0000	1.0000	1.0000	1.0000	1.0000

Figure 1 — Full-Cohort Experimental Results

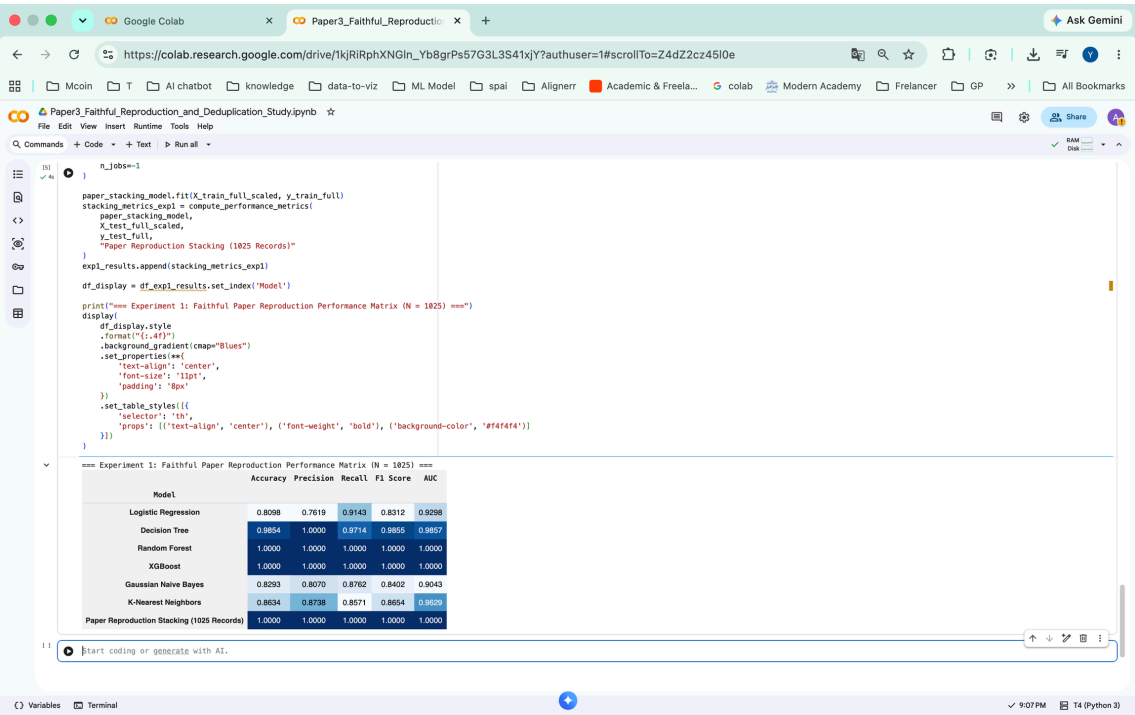


figure 1: Full-Cohort Experimental Results (N = 1,025) — Random Forest, XGBoost, and Stacking Ensemble achieved perfect classification (Accuracy = 1.0000, AUC

Random Forest, XGBoost, and the Stacking Ensemble all achieved 100.00% accuracy and an AUC of 1.0000 on the held-out test partition. The Decision Tree achieved 98.54% accuracy. Logistic Regression and Gaussian Naive Bayes performed at more moderate levels (80.98% and 82.93% respectively), consistent with simpler linear or probabilistic decision boundaries.

4.2 Experiment 2 — Deduplicated Cohort Analysis (N = 302)

The second experiment used the 302 unique observations obtained after exact-record deduplication. The performance profile changed considerably.

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	0.803	0.800	0.848	0.823	0.871
Decision Tree	3	0	5	5	2
Random Forest	0.803	0.818	0.818	0.818	0.801
XGBoost	3	2	2	2	9
Gaussian Naive Bayes	0.754	0.764	0.787	0.776	0.858
K-Nearest Neighbors	1	7	9	1	8
Stacking Ensemble	0.721	0.735	0.757	0.746	0.832
	3	3	6	3	3
	0.786	0.833	0.757	0.793	0.884
	9	3	6	7	2
	0.786	0.777	0.848	0.811	0.837
	9	8	5	6	7
	0.754	0.764	0.787	0.776	0.862
	1	7	9	1	6

Figure 2 — Deduplicated-Cohort Results

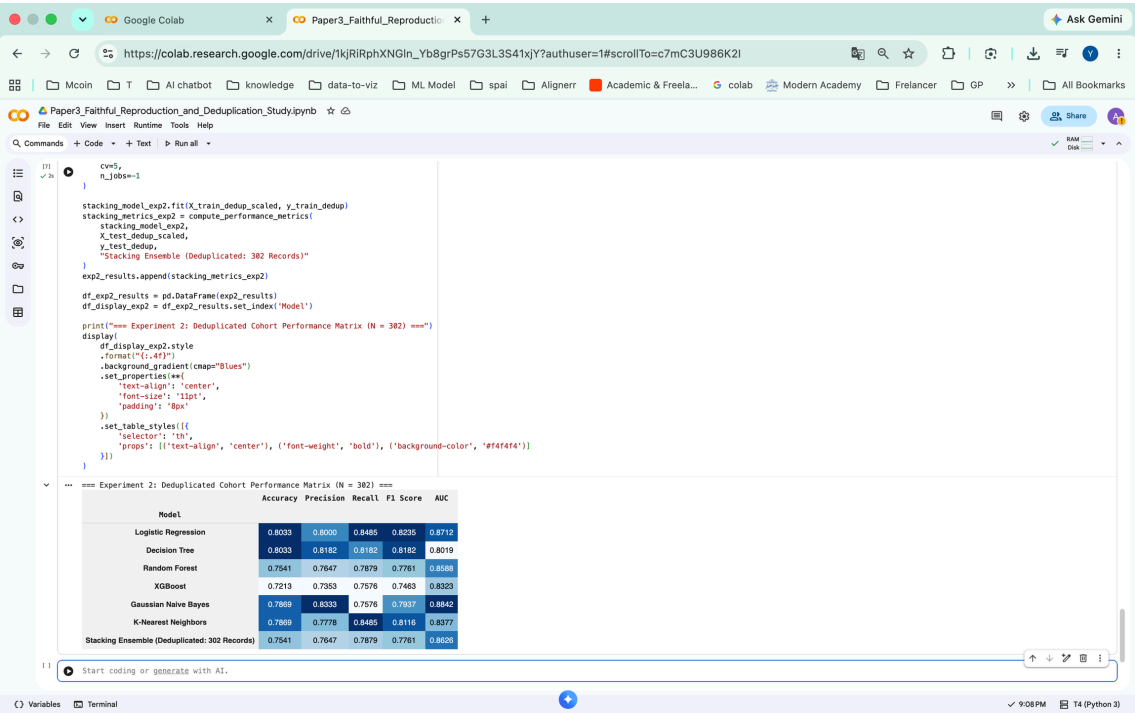


Figure 2: Deduplicated-Cohort Results (N = 302) — Logistic Regression achieves the highest accuracy (80.33%); Gaussian Naive Bayes achieves the highest AUC at 0.8842.

Logistic Regression achieved the highest accuracy at 80.33%, while Gaussian Naive Bayes achieved the highest AUC at 0.8842. The more complex ensemble models performed below the simpler linear baseline, highlighting the effect of reduced sample size on complex learners.

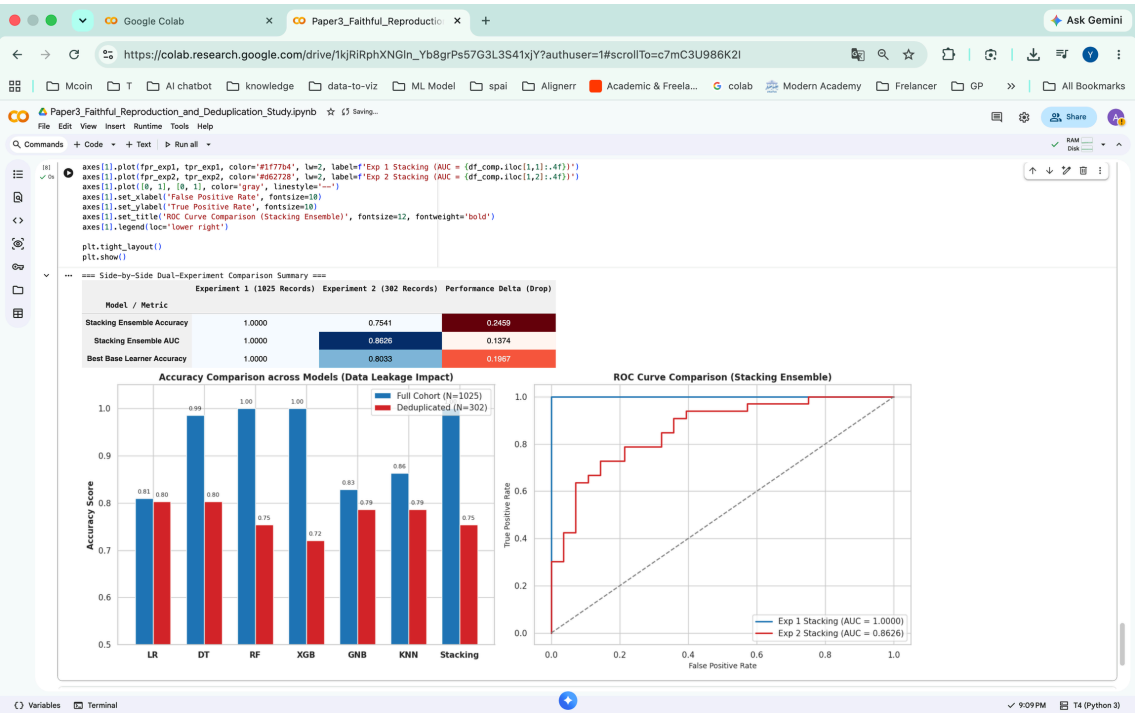
5. Comparative Diagnostics and Contamination Analysis

5.1 Side-by-Side Dual-Experiment Comparison

Full Cohort (N = 1,025)

Metric		Deduplicated (N = 302)	Performance Change
Stacking Accuracy	1.0000	0.7541	-0.2459 (-24.59 pp)
Stacking AUC	1.0000	0.8626	-0.1374
Best Overall Accuracy	1.0000	0.8033	-0.1967 (-19.67 pp)
Best Overall AUC	1.0000	0.8842	-0.1158

Figure 3 — Dual-Experiment Comparison



The comparison demonstrates that removing exact duplicate observations substantially changed the measured predictive performance across all models and metrics. The Stacking Ensemble experienced the largest observed accuracy reduction, falling from 100.00% to 75.41%.

5.2 Comparison with the Published Paper

Source	Stacking Accuracy	Stacking AUC
Paper 3 (reported)	98.53%	0.9880
This reproduction	100.00%	1.0000
Deduplicated cohort	75.41%	0.8626

The reproduction was directionally consistent with the original paper but produced higher values. The difference may result from variations in random partitioning, software environment, preprocessing implementation, model configuration, or other experimental details.

6. Discussion

6.1 Direct Empirical Evidence of Train-Test Contamination

Metric	Value
Total test observations	205

<i>Metric</i>	<i>Value</i>
Test observations with exact copies in training	202
Direct train-test contamination rate	98.54%

The contamination rate is calculated as:

$$\text{Contamination Rate} = (202 / 205) \times 100 = 98.54\%$$

Therefore, 202 of the 205 test observations had exact duplicate counterparts in the training partition. This provides direct empirical evidence of train-test contamination caused by duplicate observations. Performance measured under such conditions should not be interpreted as evidence of genuine out-of-sample generalisation.

The analysis establishes the presence and extent of duplicate overlap; it does not demonstrate that duplication alone caused every observed performance value. Nevertheless, it provides a strong methodological explanation for the exceptionally optimistic results on the full 1,025-record cohort.

6.2 Model Behaviour on the Deduplicated Cohort

Logistic Regression achieved the highest accuracy (80.33%) and an AUC of 0.8712, indicating relative robustness compared with the more complex ensemble models under the reduced sample size.

Gaussian Naive Bayes achieved the highest AUC (0.8842), indicating strong ranking performance across positive and negative observations despite slightly lower accuracy than Logistic Regression. **Tree-based and ensemble models** all experienced substantial accuracy reductions after deduplication: Random Forest = 75.41%, XGBoost = 72.13%, Stacking Ensemble = 75.41%. These results indicate that model complexity alone does not guarantee improved performance on a relatively small clinical dataset.

7. Artefact Serialisation and Reproducibility

<i>Artefact</i>	<i>Purpose</i>
paper3_stacking_exp1_1025.joblib	Trained full-cohort Stacking Ensemble Full-
scaler_exp1_1025.joblib	cohort fitted Z-score scaler
paper3_stacking_exp2_302.joblib	Trained deduplicated-cohort Stacking
scaler_exp2_302.joblib	Ensemble Deduplicated-cohort fitted Z-
exp1_full_cohort_results.csv	score scaler Experiment 1 model results
exp2_deduplicated_results.csv	Experiment 2 model results
dual_experiment_comparison_summary.csv	Comparative experiment summary

Figure 4 — Serialised Artefacts

```

import joblib

os.makedirs('paper3_artifacts', exist_ok=True)

joblib.dump(paper_stacking_model, 'paper3_artifacts/paper3_stacking_exp1_1025.joblib')
joblib.dump(scaler_full, 'paper3_artifacts/scaler_exp1_1025.joblib')

joblib.dump(stacking_model_exp2, 'paper3_artifacts/paper3_stacking_exp2_302.joblib')
joblib.dump(scaler_dedup, 'paper3_artifacts/scaler_exp2_302.joblib')

df_exp1_results.to_csv('paper3_artifacts/exp1_full_cohort_results.csv', index=False)
df_exp2_results.to_csv('paper3_artifacts/exp2_deduplicated_results.csv', index=False)
df_comp.to_csv('paper3_artifacts/dual_experiment_comparison_summary.csv', index=False)

print("[SUCCESS] All model artifacts, scalers, and performance evaluation metrics serialized to 'paper3_artifacts/'.")

[SUCCESS] All model artifacts, scalers, and performance evaluation metrics serialized to 'paper3_artifacts/'.

```

Figure 4: Serialised Artefacts — trained pipelines (.joblib) and result tables (.csv) as stored in the project repository for full reproducibility.

All trained pipelines and experimental outputs were serialised to support reproducibility and inspection. These artefacts allow the experimental results to be verified and reproduced from the submitted project files without re-running the full training procedure.

8. Limitations and Recommendations

8.1 Study Limitations

- 1. Reduced sample size after deduplication.** Exact-record deduplication reduced the dataset from 1,025 to 302 unique observations, increasing uncertainty and variance in performance estimates.
- 2. Single train-test split.** A single stratified 80/20 split with fixed seed 42 was used. Repeated stratified splits or repeated cross-validation could provide stronger evidence of performance stability.
- 3. Lack of explicit patient identifiers.** The duplicate analysis was based on exact feature tuples rather than patient IDs. Identical records cannot automatically be interpreted as belonging to the same patient.
- 4. Fixed model configurations.** Models were evaluated at specified baseline settings rather than through extensive hyperparameter optimisation.

5. No external clinical validation. Independent external datasets and clinical validation would be required before drawing conclusions about real-world diagnostic generalisation.

8.2 Recommendations for Medical Machine Learning Research

1. Perform duplicate audits before dataset splitting. Exact and, where appropriate, near-duplicate observations should be investigated before creating training and test partitions.
2. Use group-aware partitioning when repeated observations exist. Methods such as GroupKFold should be considered when patient IDs or other grouping identifiers are available.
3. Include strong baseline models. Linear models such as Logistic Regression should be retained as meaningful baselines, particularly when working with small clinical datasets.
4. Investigate unusually high performance. Near-perfect performance on small medical datasets should trigger additional validation and data-integrity checks.
5. Use independent validation datasets. External validation is essential for assessing whether a model generalises beyond the source dataset.

9. Conclusion

This study reproduced the machine learning methodology of Paper 3 and extended the evaluation with an empirical investigation of duplicate observations and train-test contamination.

Using the complete 1,025-observation dataset, the reproduction produced exceptionally high performance. Random Forest, XGBoost, and the Stacking Ensemble achieved 100.00% accuracy and an AUC of 1.0000, compared with the original paper's reported Stacking Ensemble performance of 98.53% accuracy and 0.9880 AUC. The dataset contained 723 duplicate observations, leaving 302 unique observations after exact-record deduplication.

The most significant finding came from the duplicate index-overlap analysis: 202 of the 205 test observations — corresponding to 98.54% of the test set — had exact duplicate counterparts in the training partition. This provides direct empirical evidence of train-test contamination in the original evaluation setting and offers a strong methodological explanation for the unusually high observed performance.

When the same model suite was evaluated on the deduplicated cohort, the Stacking Ensemble accuracy decreased from 100.00% to 75.41%, while Logistic Regression achieved the highest accuracy at 80.33% and Gaussian Naive Bayes achieved the highest AUC at 0.8842.

Overall, the findings demonstrate that exceptionally high medical machine learning performance should be interpreted cautiously when duplicate observations are present. Rigorous duplicate auditing, appropriate partitioning strategies, and independent validation are essential for obtaining more reliable estimates of model generalisation and clinical usefulness.

References

- [1] M. Bhagat, A. Sharma, and P. Agarwal, "An efficient stacking-based ensemble technique for early heart attack prediction," **Multimedia Tools and Applications**, vol. 84, pp. 36351–36375, 2025, doi: 10.1007/s11042-024-19293-7.
- [2] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in **14th Int. Joint Conf. on Artificial Intelligence (IJCAI)**, 1995, pp. 1137–1143.
- [3] L. Breiman, "Random forests," **Machine Learning**, vol. 45, no. 1, pp. 5–32, 2001.
- [4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in **Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining**, 2016, pp. 785–794.

Generative AI Acknowledgement Statement

In accordance with Deakin University academic integrity requirements, Generative AI tools were used for limited language and presentation support, including brainstorming, report organisation, formatting, and refinement of grammar and academic language. All technical implementation, dataset preprocessing, machine learning experiments, code execution, evaluation audits, result verification, and analytical conclusions were independently completed by the author.